

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? i.e. working parents of 0-5 y.o. kids The main customers for the malaria detection system are hospitals, diagnostic laboratories, and healthcare centers that require quick and accurate diagnosis of malaria infections. Medical professionals such as doctors, pathologists, and laboratory technicians will use the system to analyze blood smear images and detect malaria parasites. In addition, rural health clinics and government healthcare organizations can benefit from this technology because they often lack experienced specialists for manual diagnosis. Researchers working in medical imaging and artificial intelligence can also use this system for further study and improvement of automated disease detection	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices Customers may face several limitations when adopting new diagnostic technologies. Many healthcare facilities, particularly in rural areas, lack advanced laboratory equipment and skilled professionals. Financial constraints may also prevent hospitals from investing in expensive medical technologies. In addition, limited computing resources or internet connectivity in some regions can affect the implementation of digital healthcare solutions. Therefore, the malaria detection system should be simple, cost-effective, and easy to use so that it can be adopted by a wide range of healthcare organizations.	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking Currently, malaria detection is commonly performed through manual microscopy, rapid diagnostic tests (RDTs), and laboratory testing methods. In manual microscopy, trained experts examine stained blood smear images under a microscope to identify malaria parasites. While this method can be accurate, it is time-consuming and requires significant expertise. Rapid diagnostic tests provide quicker results but may not always be fully reliable. Laboratory-based testing is another option, but it can be expensive and time-intensive. These limitations highlight the need for automated systems that can improve speed and accuracy in malaria diagnosis.	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides. The primary task that customers need to perform is the identification of malaria parasites in blood cell images. Traditionally, this process is carried out manually by examining microscopic blood smear slides, which requires trained experts and a considerable amount of time. However, manual diagnosis can be slow and may lead to human errors, especially when large numbers of samples must be examined. Healthcare organizations therefore require a faster, more efficient, and reliable method to detect malaria infections so that patients can receive timely treatment.	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? i.e. customers have to do it because of the change in regulations. The main root cause of the malaria detection problem is the reliance on manual diagnostic methods that require skilled professionals and considerable time. Many regions with high malaria prevalence lack trained pathologists and advanced diagnostic equipment. As a result, the diagnosis process may be slow, and errors can occur due to fatigue or limited expertise. The increasing number of malaria cases further increases the workload for healthcare professionals, making it difficult to analyze every sample carefully using traditional methods.	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace) Healthcare professionals typically follow traditional procedures when diagnosing malaria. They collect blood samples from patients, prepare thin blood smear slides, and examine them manually under microscopes. In some cases, samples are sent to specialized laboratories for further analysis and confirmation. Medical professionals may also rely on diagnostic kits and laboratory reports to identify malaria infections. Additionally, doctors and researchers often search online for medical technologies, tools, and research studies that can help improve diagnostic accuracy and efficiency.	
Focus on J&P, tap into BE, understand RC	3. TRIGGERS TR What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news. The increase in malaria cases and the need for faster diagnosis trigger hospitals to search for better solutions. Heavy workloads for doctors and the availability of artificial intelligence technology also encourage healthcare organizations to adopt automated diagnostic systems.	10. YOUR SOLUTION SI If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. The proposed solution is an artificial intelligence-based malaria detection system developed using deep learning techniques. The system uses a pre-trained MobileNetV2 model to analyze microscopic blood cell images and classify them as either parasitized or uninfected. A web application built with the Flask framework allows users to upload blood cell images and receive instant predictions along with confidence scores. This automated system helps reduce diagnosis time, minimizes human error, and supports healthcare professionals in making faster and more accurate decisions, ultimately improving patient care and malaria treatment outcomes.	8.CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. Customers interact with both online and offline channels while performing their tasks. Online channels include medical software systems, healthcare technology websites, research databases, and professional communities where doctors and researchers share knowledge and discuss new technologies. Offline channels include hospitals, clinics, diagnostic laboratories, medical conferences, and training programs. These channels help healthcare professionals learn about new diagnostic tools and technologies that can improve their work efficiency.	Identify strong TR & EM
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? i.e. lost, insecure > confident, in control - use it in your communication strategy & design. Before using an automated malaria detection system, healthcare professionals often feel stressed and overwhelmed due to the large number of blood samples that must be analyzed manually. The manual examination process can be tiring and increases the risk of diagnostic errors. Patients may also feel anxious and worried while waiting for their diagnosis results. After implementing an AI-based malaria detection system, healthcare professionals can work more efficiently and confidently because the system assists them in analyzing images quickly and accurately. Patients benefit from faster diagnosis and earlier treatment, which improves their confidence and overall healthcare experience.			